**Week-5**
Mathematics for Data Science - 2
Rank and Nullity of a matrix
**Practice Assignment**

1. Let $A$ and $B$ be square matrices of order 2. Which of the following options is true?
   - ◯ Option 1: $nullity(AB) = nullity(A) + nullity(B)$
   - ◯ Option 2: $nullity(AB) = nullity(A)nullity(B)$
   - ◯ Option 3: $nullity(A - B) = nullity(A) - nullity(B)$
   - ◯ Option 4: $rank(A + B) = rank(A) + rank(B)$
   - ◯ **Option 5:** None of the above

## Solution:

**Option 1:**

Take two matrices $A$ and $B$ as follows:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\implies AB = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

Observe that $nullity(A) = 1$ and $nullity(B) = 1$. But $nullity(AB) = 1$. Clearly, $nullity(AB) \neq nullity(A) + nullity(B)$. Hence, this option is **incorrect**.

**Option 2:**

Similar to option 1, take two matrices $A$ and $B$ as follows:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

$$\implies AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Observe that $nullity(A) = 1$ and $nullity(B) = 1$. But $nullity(AB) = 2$. Clearly, $nullity(AB) \neq nullity(A)nullity(B)$. Hence, this option is also **incorrect**.

**Option 3:**

Take two matrices $A$ and $B$ as follows:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 3 & 4 \end{bmatrix}$$

$$\implies A - B = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

Observe that $nullity(A) = 0$ and $nullity(B) = 1$. But $nullity(A - B) = 1$. Clearly, $nullity(A - B) \neq nullity(A) - nullity(B)$. Hence, this option is **incorrect**.

**Option 4:**

Take two matrices $A$ and $B$ as follows (or any two non-zero matrices whose sum equals the zero matrix) :

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & -2 \\ -1 & -2 \end{bmatrix}$$

$$\implies A + B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Observe that $rank = 1$ and $rank(B) = 1$. But $rank(A+B) = 0$. Clearly, $rank(A+B) \neq rank(A) + rank(B)$. Hence, this option is **incorrect**.

**Option 5:**

Since none of the above options are correct, this option is **correct**.

2. A molecule is composed of atoms. A molecule of Ethylene, with the chemical formula $C_2H_4$, consists of two Carbon atoms and four Hydrogen atoms. A molecule of Oxygen, with the formula $O_2$, consists of two Oxygen atoms. Note that Carbon, Hydrogen and Oxygen are denoted by the letters $C, H$, and $O$ respectively in the formula. When Ethylene comes in contact with Oxygen ($O_2$); Carbon dioxide ($CO_2$) and water ($H_2O$) are produced as the products of the chemical reaction . The equation corresponding to the chemical reaction (R) is given below

$$x_1C_2H_4 + x_2O_2 \longrightarrow x_3CO_2 + x_4H_2O. \qquad \ldots \text{(R)}$$

To balance the chemical equation we have to choose $x_1, x_2, x_3,$ and $x_4$ such that both sides have the same number of carbon atoms on each side, the same number of hydrogen atoms on each side, and the same number of oxygen atoms on each side.
Note: An example to write the system of linear equations for balancing the chemical equation is the following :

$$x_1C + x_2O \longrightarrow x_3CO_2$$

corresponding system of linear equations is:

$$x_1 = x_3$$
$$x_2 = 2x_3$$

Consider the system of linear equations obtained for balancing the chemical equation (R) to answer the question.
Which of the following statements are true?

- ◯ Option 1: The nullity of the matrix corresponding to this system is 0.
- ◯ **Option 2:** The nullity of the matrix corresponding to this system is 1.
- ◯ Option 3: $\{(1, 3, 1, 1)\}$ is a basis of the null space of the matrix corresponding to this system.
- ◯ **Option 4:** There are an infinite number of ways to balance the chemical equation (R).
- ◯ Option 5: There is a unique way to balance the chemical equation (R).

## Solution:

The system of linear equations for balancing the chemical equation R can be be given as follows:

$$\left.\begin{aligned} 2x_1 &= x_3 \\ 4x_1 &= 2x_4 \\ 2x_2 &= 2x_3 + x_4 \end{aligned}\right\} \implies \begin{aligned} 2x_1 - x_3 &= 0 \\ 2x_1 - x_4 &= 0 \\ 2x_2 - 2x_3 - x_4 &= 0 \end{aligned}$$

The matrix representation of the above system can be given as follows:

$$A = \begin{bmatrix} 2 & 0 & -1 & 0 \\ 2 & 0 & 0 & -1 \\ 0 & 2 & -2 & -1 \end{bmatrix}$$

By reducing the matrix to its reduced row echelon form, we get:

$$R = \begin{bmatrix} 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

The number of independent variables is 1, hence the nullity of the system if 1. Hence, **Option 2** is correct, and option 1 is incorrect.

Since there is an independent variable, there are infinitely many ways of balancing the above equation. Hence, **Option 4** is correct, and option 5 is incorrect.

For option 2, if the given set was a basis of the null space, then the following should hold true:

$$A \cdot \begin{bmatrix} 1 \\ 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

But we have

$$\begin{bmatrix} 2 & 0 & -1 & 0 \\ 2 & 0 & 0 & -1 \\ 0 & 2 & -2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Thus, the given set cannot be a basis for the null space and hence this option is incorrect.

3. Which of the following option(s) is(are) true?

   ◯ **Option 1:** If $A_{2\times 3}$ is a non zero matrix, then nullity of the matrix $\leq 2$.

   ◯ **Option 2:** If $A_{3\times 2}$ is a non zero matrix, then nullity of the matrix $\leq 1$.

   ◯ **Option 3:** Let $A$ and $B$ be two square matrices of order 3, if nullity of matrix $AB$ is 0, then nullity of matrix $A$ is also zero.

   ◯ **Option 4:** Let $A$ and $B$ be two square matrices of order 3, if nullity of matrix $AB$ is 0, then nullity of matrix $B$ is also zero.

## Solution:

For any given $m \times n$ matrix $A$, $rank(A) + nullity(A) = n$.

**Option 1:**

If $A$ is a $2 \times 3$ matrix, then $rank(A) + nullity(A) = 3$. Also we are given that $A$ is a non- zero matrix, thus $rank(A) \geq 1$. This implies nullity$(A) \leq 2$.

Hence, this option is **correct**.

**Option 2:**

If $A$ is a $3 \times 2$ matrix, then $rank(A) + nullity(A) = 2$. Since $A$ is a non zero matrix, $rank(A) \geq 1$ and thus $nullity(A) \leq 1$.

Hence, this option is also **correct**.

**Options 3 and 4:**

If $nullity(AB) = 0$, then $AB$ is an invertible matrix. Then $det(AB) \neq 0$ and thus $det(A)det(B) \neq 0$. Thus, both $A$ and $B$ have non-zero determinant. Hence both the matrices are invertible. That is, both the matrices have $nullity = 0$. Hence, both these options are **correct**.

4. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear transformation such that $T(1,0,0) = (1,1,1), T(1,1,0) = (0,0,1)$ and $T(1,1,1) = (0,1,1)$. Then which of the following option is/ are true?

   ◯ Option 1: $T(x,y,z) = (x-y, x-y+z, y)$

   ◯ **Option 2:** $T(x,y,z) = (x-y, x-y+z, x)$

   ◯ Option 3: $T(x,y,z) = (x+y, x-y+z, x)$

   ◯ **Option 4:** $T$ is one-one.

   ◯ Option 5: $T$ is not onto.

## Solution:

**Options 1, 2, and 3:**

Observe that $\{(1,0,0),(1,1,0),(1,1,1)\}$ is a basis for $\mathbb{R}^3$. Thus, any vector $(x,y,z) \in \mathbb{R}^3$ can be represented as a linear combination of these vectors using scalar multiples $a, b, c$ as follows:

$$\begin{aligned}(x,y,z) &= a \cdot (1,0,0) + b \cdot (1,1,0) + c \cdot (1,1,1) \\ &= (a+b+c, b+c, c)\end{aligned} \tag{1}$$

This gives us the following values for $a, b,$ and $c$:

$$\left.\begin{aligned} a+b+c &= x \\ b+c &= y \\ c &= z \end{aligned}\right\} \implies \begin{aligned} a &= x-y \\ b &= y-z \\ c &= z \end{aligned} \tag{2}$$

The transformation of any vector can be given as follows:

$$\begin{aligned}T(x,y,z) &= a \cdot T(1,0,0) + b \cdot T(1,1,0) + c \cdot T(1,1,1) \\ &= a \cdot (1,1,1) + b \cdot (0,0,1) + c \cdot (0,1,1) \\ &= (a, a+c, a+b+c)\end{aligned} \tag{3}$$

Substituting the values from (2) in (3), we get:

$$T(x,y,z) = (x-y, x-y+z, x) \tag{4}$$

Thus, **option 2** is correct.

**Option 4:**

We know that a linear transformation is one-to-one if and only if $T(x, y, z) = 0$ implies $(x, y, z) = 0$. Let $T(x, y, z) = (0, 0, 0)$. Then we have the following:

$$\begin{aligned} T(x, y, z) &= (x - y, x - y + z, x) \\ \implies (x - y, x - y + z, x) &= (0, 0, 0) \end{aligned}$$

This gives us the following system of equations:

$$\left.\begin{aligned} x - y &= 0 \\ x - y + z &= 0 \\ x &= 0 \end{aligned}\right\} \implies \begin{aligned} x &= 0 \\ y &= 0 \\ z &= 0 \end{aligned}$$

Thus, $T(x, y, z) = (0, 0, 0) \implies (x, y, z) = (0, 0, 0)$. Hence, we can say that the given transformation is one-to-one. This option is **correct**.

**Option 5:**

If for any vector $(p, q, r)$, there exist $(x, y, z)$ such that $T(x, y, z) = (p, q, r)$, then the linear transformation $T$ is onto. We have:

$$\left.\begin{aligned} T(x, y, z) &= (p, q, r) \\ \implies (x - y, x - y + z, x) &= (p, q, r) \end{aligned}\right\} \implies \begin{aligned} x &= r \\ y &= r - p \\ z &= q - p \end{aligned}$$

Thus, for every output $(p, q, r)$, we have an input $(x, y, z)$, which makes $T$ onto. Thus, this option is **incorrect**.

5. Consider a system of linear equations defined as follows:

$$\begin{aligned} x + y - z + w &= 0 \\ 2x + y + 3z - w &= 0 \\ 3x + 2y + 2z &= 0 \end{aligned}$$

If $A$ is the coefficient matrix of the system of linear equations, then which of the following options is/are true?

- ◯ **Option 1:** Null space of $A = span\{(-4, 5, 1, 0), (2, -3, 0, 1)\}$
- ◯ **Option 2:** Null space of $A = span\{(-6, 8, 1, -1), (-4, 5, 1, 0)\}$

◯ Option 3: Null space of $A = span\{(-3, 5, 1, 0), (2, -3, 0, 1)\}$

◯ Option 4: Null space of $A = span\{(-4, 5, 1, -1), (6, 7, 2, 1)\}$

## Solution:

We know that $A$ is the coefficient matrix of the system. Thus, we have:

$$A = \left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 2 & 1 & 3 & -1 & 0 \\ 3 & 2 & 2 & 0 & 0 \end{array}\right]$$

Reducing the matrix to reduced row echelon form $R$ gives us the following:

$$R = \left[\begin{array}{cccc|c} 1 & 0 & 4 & -2 & 0 \\ 0 & 1 & -5 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

**Options 1 and 3:**

Observe that $z$ and $w$ are the free variables. Let $z = 0$ and $w = 1$. This gives us the following system:

$$\left.\begin{array}{l} x + 0y + (4 \cdot 0) + (-2 \cdot 1) = 0 \\ 0x + y + (-5 \cdot 0) + (3 \cdot 1) = 0 \end{array}\right\} \implies \begin{array}{l} x = 2 \\ y = -3 \end{array}$$

Thus, we get $span(2, -3, 0, 1) \in N(A)$. Similarly, substitute $z = 1$ and $w = 0$ in the matrix $R$, we get:

$$\left.\begin{array}{l} x + 0y + (4 \cdot 1) + (-2 \cdot 0) = 0 \\ 0x + y + (-5 \cdot 1) + (3 \cdot 0) = 0 \end{array}\right\} \implies \begin{array}{l} x = -4 \\ y = 5 \end{array}$$

Thus, we get $span(-4, 5, 1, 0) \in N(A)$. This gives us the following:

$$N(A) = span\,\{(-4, 5, 1, 0), (2, -3, 0, 1)\}$$

Hence, **option 1** is correct, and option 3 is incorrect.

**Option 2:**

We already know that $span(-4, 5, 1, 0) \in N(A)$. Thus, we need to check whether $(-6, 8, 1, -1) \in N(A)$ or not. Substitute the values of the free variables in $R$, we get the following:

$$\left.\begin{array}{l} x + 0y + (4 \cdot 1) + (-2 \cdot -1) = 0 \\ 0x + y + (-5 \cdot 1) + (3 \cdot -1) = 0 \end{array}\right\} \implies \begin{array}{l} x = -6 \\ y = 8 \end{array}$$

Thus, we get $(-6, 8, 1, -1) \in N(A)$. Hence, this option is also correct.

**Option 2:**

Similar to option 4, we can prove that this set does not form the null space.

## Numerical Answer Type (NAT)

6. If nullity of a matrix $A_{5\times 7}$ is 2, then the find the rank of the matrix $A_{5\times 7}$. [Ans: 5]

**Solution:**

We are given that the $nullity(A) = 2$. Since $A$ is a $5 \times 7$ matrix, we know the following:

$$\begin{aligned} rank(A) + nullity(A) &= 7 \\ \implies rank(A) &= 5 \end{aligned}$$

Thus the correct answer is **5**.

7. Let $A$ be non-zero matrix of order $3 \times 1$. What is the rank of the matrix $AA^T$? [Ans: 1]

## Solution:

We are given that $A$ is a non-zero matrix of order $3 \times 1$. Observe that $Rank(A) = 1$. We know that $Rank(A) = Rank(A^T)$. We have the following:

$$\begin{aligned} Rank(A) &= Rank(A^T) = 1 \\ Rank(AA^T) &\leq min\{Rank(A), Rank(B)\} \\ \implies Rank(AA^T) &\leq 1 \end{aligned}$$

Since $A$ is a non-zero matrix, there will be at least one non-zero entry in the resultant $AA^T$. Hence, $Rank(AA^T)$ cannot be zero. Thus, we have $Rank(AA^T) = 1$.

## Comprehension Type Questions:

Ayesha, Pritha, Sabya, and Wang went on a trip to Manali and Kasol. Accommodation costs ₹1500 per day in Manali and ₹800 per day in Kasol. The total food cost is ₹2000 per day in Manali and ₹1200 per day in Kasol. They plan to spend 2 days in Manali and 2 days in Kasol. The first and second rows of the Table M2W5P1 shows the percentage of contribution by each of them for the accomodation at Manali and Kasol, respectively. Similarly, the first and the second row of the Table M2W5P3 shows the percentage of contribution by each of them for the food at Manali and Kasol, respectively.

**Table for Accommodation cost:**

| | Ayesha | Pritha | Sabya | Wang |
|---|---|---|---|---|
| Manali | $x_1\%$ | $x_2\%$ | $x_3\%$ | $x_4\%$ |
| Kasol | $y_1\%$ | $y_2\%$ | $y_3\%$ | $y_4\%$ |

Table: M2W5P1

**Table for cost of food:**

| | Ayesha | Pritha | Sabya | Wang |
|---|---|---|---|---|
| Manali | $v_1\%$ | $v_2\%$ | $v_3\%$ | $v_4\%$ |
| Kasol | $w_1\%$ | $w_2\%$ | $w_3\%$ | $w_4\%$ |

Table: M2W5P2

Suppose $T(x, y)$ denotes the contribution of a person for accommodation per day, where the first variable $x$ denotes the percentage of contribution by that person for accommodation in Manali and the second variable $y$ denotes the percentage of contribution by that person for accommodation in Kasol. (i.e., if $a$ and $b$ denote the costs for accommodation per day at Manali and Kasol, respectively, then $T(x, y) = \frac{1}{100}(ax + by)$). Similarly, $T'(v, w)$ denotes the contribution by a person for food per day, where the first variable $v$ denotes the percentage of contribution for food in Manali and the second variable $w$ denotes the percentage of contribution for food in Kasol. Answer the following questions based on the given information.

8. Choose the set of correct options. (MSQ)

   ◯ Option 1: $T(x, y) = 30x + 16y$

   ◯ **Option 2:** $T(x, y) = 15x + 8y$

   ◯ **Option 3:** $T'(v, w) = 20v + 12w$

◯ Option 4: $T'(v, w) = 40v + 24w$

## Solution:

We are given that if $a$ and $b$ denote the costs for accommodation per day at Manali and Kasol, respectively, then $T(x, y) = \frac{1}{100}(ax + by))$. That is the accommodation costs are as follows:

**Accommodation Costs:**

₹1500 Per Day in Manali

₹800 Per Day in Kasol

Thus we have equation for the contribution for accommodation as follows:

$$T(x, y) = \frac{1}{100}(1500x + 800y)$$
$$T(x, y) = 15x + 8y$$

We are also given that $T'(v, w)$ denotes the contribution by a person for food per day. That is, $T'(v, w) = \frac{1}{100}(cx + dy)$. We have the following costs:

**Total Food Costs:**

₹2000 Per Day in Manali

₹1200 Per Day in Kasol

Thus, we have the equation for the contribution for the food as follows:

$$T'(v, w) = \frac{1}{100}(2000v + 1200w)$$
$$T'(v, w) = 20v + 12w$$

Hence, the correct options are **Option 2** and **Option 3**.

9. Choose the set of correct options. (MSQ)

◯ **Option 1:** Suppose Ayesha contributes for herslef and also on behalf of Sabya. Then the contribution per day by Ayesha is given by $T(x_1 + x_3, y_1 + y_3)$ and $T'(v_1 + v_3, w_1 + w_3)$, for accommodation and food, respectively.

◯ **Option 2:** The total contribution (for the whole trip) for the accommodation by Pritha is given by $2T(x_2, y_2)$, which is equal to $T(2x_2, 2y_2)$.

◯ Option 3: The total contribution (for the whole trip) for the accommodation by Pritha is given by $2T(x_2, y_2)$, which is not equal to $T(2x_2, 2y_2)$.

◯ Option 4: Suppose Pritha contributes for herself and also on behalf of Wang for food. Then the contribution per day by Pritha for food is given by $T'(v_2, w_2) + T'(v_4, w_4)$, which is not equal to $T'(v_2 + v_4, w_2 + w_4)$.

○ **Option 5:** Suppose Pritha contributes for herself and also on behalf of Wang for food. Then the contribution per day by Pritha for food is given by $T'(v_2, w_2) + T'(v_4, w_4)$, which is equal to $T'(v_2 + v_4, w_2 + w_4)$.

## Solution:

### Option 1:

$T(x, y)$ denotes the contribution of a person for accommodation per day, where $x$ is the percentage contribution for the accommodation in Manali, and $y$ is the percentage contribution for the accommodation in Kasol. From the table M2W5P1, We have the percentage contributions of Ayesha and Sabya as follows:

$$\begin{gathered}\text{Accommodation contribution of Ayesha at Manali} = x_1\% \\ \text{Accommodation contribution of Ayesha at Kasol} = y_1\% \\ \text{Accommodation contribution of Sabya at Kasol} = x_3\% \\ \text{Accommodation contribution of Sabya at Kasol} = y_3\%\end{gathered}$$

If Ayesha contributes for herself and for Sabya, then the total accommodation contribution for Manali $= (x_1 + x_3)\%$. Similarly, for Kasol it will be $= (y_1 + y_3)\%$. Thus, we have the total accommodation contribution of Ayesha as follows:

$$T(x, y) = (x_1 + x_3, y_1 + y_3)$$

Similarly, we can find the total food contribution as well. It will come out to be $T'(v_1 + v_3, w_1 + w_3)$. Hence, this option is **correct**.

### Options 2 and 3:

Observe that $T(x, y)$ is a linear transformation. The total contribution for accommodation by Pritha for one day is $T(x_2, y_2)$. Thus, the contribution of Pritha for two days' of accommodation is $2T(x_2, y_2)$. Because $T(x, y)$ is a linear transformation, the following holds true:

$$2T(x_2, y_2) = T(2x_2, 2y_2)$$

Hence **Option 2** is correct, while option 3 is incorrect.

### Options 4 and 5:

Observe that $T'(v, w)$ is also a linear transformation. If Pritha contributes for her food and Wang's food, then the total contribution by Pritha for food would be given by the following:

$$T'(v_2 + v_4, w_2 + w_4) = T'(v_2, w_2) + T'(v_3, w_4)$$

Hence **Option 5** is correct, while option 4 is incorrect.

10. The total contribution (accommodation and food) per day can be denoted by: (MSQ)

- ◯ Option 1: $f(x, y, v, w) = T(x, y) + T'(v, w)$ which is not a linear mapping.
- ◯ **Option 2:** $f(x, y, v, w) = T(x, y) + T'(v, w)$ which is a linear mapping.
- ◯ Option 3: $f(2x, 2y, 2v, 2w) = 2^4 f(x, y, v, w)$.
- ◯ **Option 4:** $f(2x, 2y, 2v, 2w) = 2f(x, y, v, w)$.

## Solution:

The total contribution is given by the following: $f(x, y, v, w) = T(x, y) + T'(v, w)$.

Consider the following:

$f(x_1 + x_2, y_1 + y_2, v_1 + v_2, w_1 + w_2)$

$$\begin{aligned}
&= T(x_1 + x_2, y_1 + y_2) + T'(v_1 + v_2, w_1 + w_2) \\
&= T(x_1, y_1) + T(x_2, y_2) + T'(v_1, w_1) + T'(v_2, w_2) \\
&= T(x_1, y_1) + T'(v_1, w_1) + T(x_2, y_2) + T'(v_2, w_2) \\
&= f(x_1, y_1, v_1, w_1) + f(x_2, y_2, v_2, w_2)
\end{aligned}$$

Also consider the following:

$$\begin{aligned}
f(cx, cy, cv, cw) &= T(cx, cy) + T'(cv, cw) \\
&= cT(x, y) + cT'(v, w) \\
&= c(T(x, y) + T'(v, w) \\
&= cf(x, y, v, w)
\end{aligned}$$

Hence, $f(x, y, v, w)$ is a linear mapping and $f(2x, 2y, 2v, 2w) = 2f(x, y, v, w)$. Thus, the **Option 2** and **Option 4** are correct.